

Application Layer

1. Introduction to the Application Layer

The **Application Layer** is the **topmost layer (Layer 7)** of the OSI model.

It provides **network services directly to user applications** and acts as an **interface between users and the network**.

Application Layer allows **software applications to communicate over a network**.

Examples of Applications Using This Layer

- Web browsers (Chrome, Firefox)
- Email applications (Gmail, Outlook)
- File transfer programs
- Remote login applications

These applications use **application layer protocols** to communicate with servers.

Position of Application Layer is 7 in OSI Model

The Application Layer interacts **directly with the user** and provides **network services to applications**.

2. Functions of the Application Layer

The Application Layer performs several important functions.

1. Network Service Access

It allows applications to **access network services**.

Example:

- Accessing a website
- Sending an email
- Uploading files

2. Data Representation

The layer ensures that **data is in a format understandable by the receiving application.**

Example:

- HTML for webpages
- JPEG for images
- ASCII for text

3. Application Communication

It allows **different applications on different systems to communicate.**

Example:

- Browser communicating with a web server.

4. Session Management

It helps **establish, manage, and terminate communication sessions** between applications. Example:

- Logging into a remote server.

3. Services Provided by the Application Layer

The Application Layer provides the following services:

1. Network Virtual Terminal

Allows a user to **log in to a remote computer.**

Example:

- **Telnet:** It is an application layer protocol used to remotely access and control another computer over a network.
- **Secure Shell (SSH):** It is a secure network protocol used to remotely access and manage computers over a network.

Difference Between Telnet and SSH

Feature	Telnet	SSH
Security	Not secure	Secure
Encryption	No encryption	Encrypted communication
Port Number	23	22
Usage Today	Mostly obsolete	Widely used

2. File Transfer Access and Management

Allows users to **access, upload, download, and manage files on remote systems.**

Example:

- File Transfer Protocol (FTP)

3. Mail Services

Allows users to **send and receive electronic mail.**

Example:

- **Simple Mail Transfer Protocol (SMTP)**

4. Directory Services

Provides **information about network resources.**

Example:

- DNS (Domain Name System)

4. Application Layer Protocols

Application Layer protocols define **rules for communication between applications**.

Protocol	Function
HTTP	Used for web communication
FTP	Used for file transfer
SMTP	Used for sending email
DNS	Converts domain names into IP addresses
DHCP	Automatically assigns IP addresses
Telnet	Remote login

5. Client–Server Model

Most Application Layer services follow the **Client–Server architecture**. The **Client–Server model** is a system where:

- **Client** requests services
- **Server** provides services

Example:

- Web browser → Client
- Web server → Server

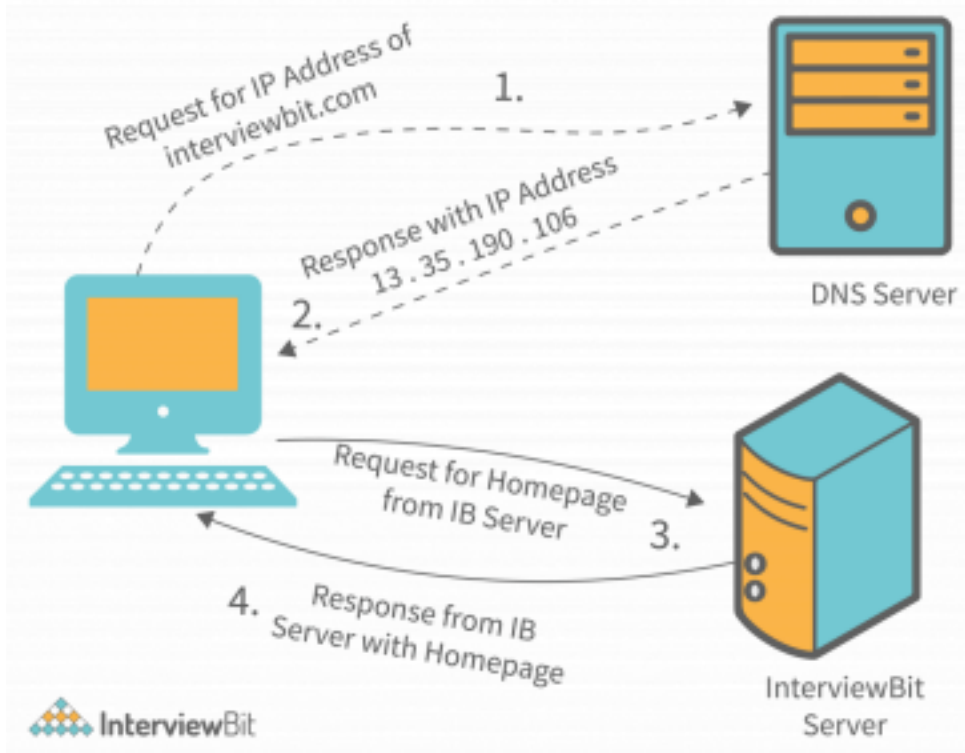
Working of Client-Server Model

1. Client sends request to server.
2. Server processes the request.
3. Server sends response to client.
4. Client displays the result.

Example:

When a user enters a website URL in a browser.

Client-Server architecture diagram



6. World Wide Web (WWW)

The World Wide Web (WWW) is a system of interconnected web pages available on the Internet. It uses the HTTP protocol for communication.

Components of WWW

1. Web Browser – software used to access websites
2. Web Server – stores and delivers webpages
3. HTTP Protocol – communication between browser and server

Example Process

4. User enters website URL.
5. Browser sends HTTP request.
6. Server processes request.
7. Server sends webpage to browser.

7. Content Distribution Network (CDN)

A Content Distribution Network (CDN) is a group of distributed servers that deliver web content based on the user's geographic location.

Purpose of CDN

1. Improve website speed
2. Reduce server load
3. Increase availability

How CDN Works

4. The user requests a webpage.
5. The request goes to the nearest CDN server.
6. CDN server delivers cached content.

Protocols

Domain Name System (DNS Protocol)

- DNS stands for Domain Name System.
- It converts domain names (like google.com) into IP addresses (like 142.250.x.x).
- Helps users access websites without remembering numbers.

How DNS Works

1. Enter Website Name

You type a domain (e.g., www.example.com).

2. Check Local Cache

Browser checks if IP is already saved.

3. Ask DNS Resolver

If not found, the request goes to the DNS resolver (ISP server).

4. Root Server

Points to the correct TLD server.

5. TLD(Top-Level Domain) Server

Directs to the domain's authoritative server.

6. Authoritative Server

Returns the correct IP address.

7. Load Website

Browser uses IP to connect and display the site.

Structure of DNS (Hierarchy)

DNS is organized in levels:

1. Root DNS Servers

- Top level
- Direct to TLD servers

2. TLD Servers (Top-Level Domain)

- Handle extensions like:

- .com
- .org

■ .net

3. Authoritative DNS Servers

- Store actual IP addresses
- Give final answer

Types of Domains

1. Generic Domains (gTLD)

- Examples:

- .com
- .org
- .net
- .edu

2. Country Code Domains (ccTLD)

- Represent countries:

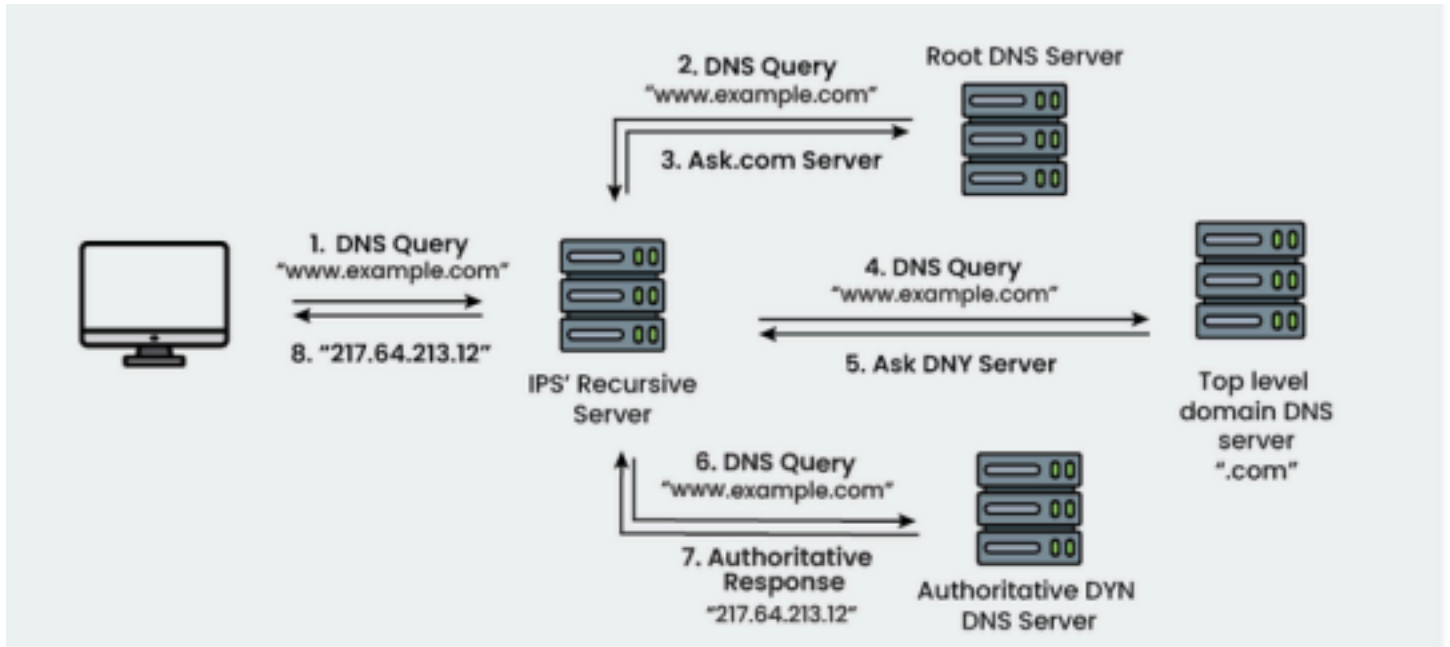
- .in (India)
- .us (USA)
- .uk (UK)
- .jp (Japan)

3. Reverse / Inverse Domains

- Used for reverse DNS lookup
- Converts IP → domain name
- Useful for security and troubleshooting

What is DNS Lookup?

DNS Lookup (DNS Resolution) = Process of converting domain name into IP address.



HTTP (HyperText Transfer Protocol)

- HTTP stands for **HyperText Transfer Protocol**.
- It is a set of rules that allows **communication between a web browser and a web server**.
- It helps load websites on your screen.

HTTP is how your browser asks for a webpage and how the server sends it

back. How HTTP Works (Step-by-Step)

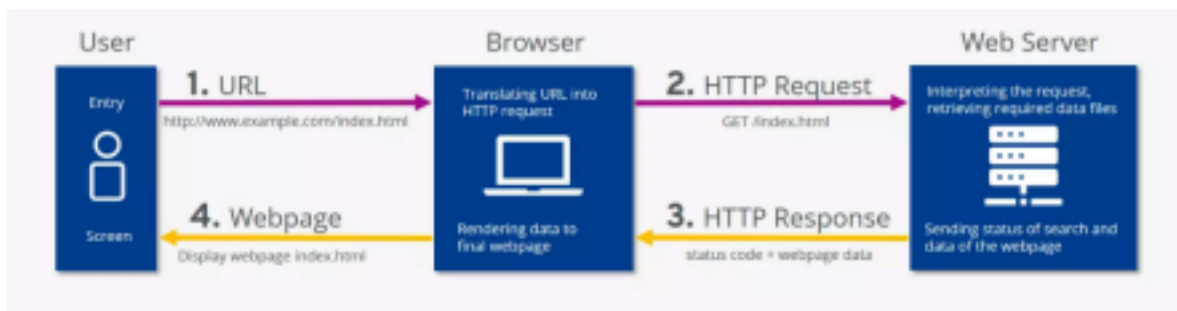
1. **Enter URL:** You type a website (e.g., google.com) in your browser. 2. **DNS**

Lookup: The browser finds the website's IP address (like finding a phone number).

3. **Send Request:** Browser sends an **HTTP request** to the server.

4. **Server Response:** Server sends back data (HTML, CSS, JS, images).

5. **Display Page:** Browser shows the webpage on your screen.



What is HyperText?

- HyperText is **text with links (hyperlinks)**.
- Clicking a link takes you to another page.
- Created using **HTML (HyperText Markup Language)**.

HTTP Request (From Browser → Server)

- **URL** – address of resource
- **Method** – action (GET, POST, etc.)
- **Headers** – extra info (browser type, etc.)

- **Body** – data (used in forms)

Example: Asking server → “Give me this webpage”

HTTP Response (From Server → Browser)

- **Status Code** – result of request
- **Headers** – info about response
- **Body** – actual content (webpage data)

HTTP Status Codes (Easy Categories)

HTTP Status codes are three-digit numbers that servers use to tell your browser what happened with the request you sent. There are different types of status codes:

1. **Informational (1xx):** These codes just give you information (e.g., 100 Continue means the request is still being processed).
2. **Successful(2xx):** These codes tell you everything went fine (e.g., 200 OK means the request was successful).
3. **Redirection(3xx):** These codes tell the browser to take additional action (e.g., 301 Moved Permanently means the requested page has moved to a new address).
4. **Client Error (4xx):** These codes indicate that there was a problem with your request (e.g., 404 Not Found means the page doesn't exist).
5. **Server Error (5xx):** These codes tell you that something went wrong on the server side (e.g., 500 Internal Server Error means the server had an issue processing the request).

Advantages of HTTP

- Simple and fast
- Low memory & CPU usage
- Supports multiple requests
- Errors can be detected easily

Disadvantages of HTTP

- Not secure (no encryption)
- Data can be intercepted
- Not ideal for mobile networks
- Less efficient compared to modern protocols

Secure version: **HTTPS (uses encryption)**

File Transfer Protocol (FTP)

- FTP stands for File Transfer Protocol
- Works at the Application Layer (OSI model)
- Used to transfer files between client and server FTP is used to

upload, download, and manage files over a network.

Key Features of FTP

- Works on client-server model
- Can transfer:
 - Text files

- Images
- Videos
- Programs
- Works between different operating systems
- Uses TCP protocol

How FTP Works

1. User connects to FTP server (login or anonymous)
2. Client sends commands (upload/download)
3. Server processes request
4. File is transferred
5. Connection remains active during session

FTP is stateful (remembers user session)

FTP Connections

1. Control Connection

- Used for commands & login
- Port: 21

2. Data Connection

- Used for file transfer
- Port: 20

FTP uses two separate connections (important point)

Common FTP Commands (Basics)

- get → Download file
- put → Upload file
- ls → List files
- cd → Change directory
- delete → Remove file
- quit → Close connection

Uses of FTP

- Uploading website files
- Backup and storage
- Transferring large files
- Remote file management

Advantages of FTP

- Fast file transfer
- Easy to use
- Supports large files
- Allows file management (rename, delete, etc.)

Disadvantages of FTP

- Not secure (no encryption)
- Passwords can be hacked

- No built-in security
- Limited modern use (replaced by SFTP/HTTPS)